Cost-Quality Ranking using Multiobjective Optimisation

A normalization-free framework to identify efficient products: low cost, high quality, and strict handling of minimum quality specifications.

Recommended idea: treat quality specification as a constraint, then rank products using Pareto dominance. This avoids forcing cost and quality into one arbitrary normalized scale.

1. Problem formulation

For each product i, you observe cost c_i and quality q_i. The natural multiobjective problem is:

$$\text{minimize } c_i$$

$$\text{maximize } q_i$$

$$\text{subject to } q_i >= q_{min}$$

This formulation keeps the original units. Cost is cost, quality is quality, and the minimum specification is not treated as something that can be compensated by low cost.

2. Feasibility first

Separate products into two groups before ranking:

Group	Condition	Treatment
Compliant	$q_i >= q_{min}$	Rank using low cost and high quality.
Non-compliant	$q_i < q_{min}$	Place below compliant products; rank by quality shortfall and cost.

3. Pareto dominance

For compliant products, product A dominates product B when A is no more expensive and no worse in quality, with at least one strict improvement:

$$c_A <= c_B \text{ and } q_A >= q_B$$

The Pareto-optimal products are those not dominated by any other product. They represent the efficient cost-quality trade-offs.

4. Example interpretation

Product	Cost	Quality	Interpretation
A	100	2.10	Pareto candidate
B	105	2.20	Pareto candidate
C	110	2.05	Dominated by A: higher cost and lower quality
D	95	1.90	Below specification, therefore infeasible

5. From Pareto set to ranking

Pareto optimisation gives a set of efficient products, not automatically a single best product. If you need an ordered ranking, use non-dominated sorting:

Layer	Meaning	Priority
Pareto front 1	Products not dominated by any other compliant product.	Best group
Pareto front 2	Products dominated only by products in front 1.	Second group
Pareto front 3+	Progressively less efficient products.	Lower groups

Within each Pareto front, apply a business tie-breaker: cost-first if quality above specification has limited value, or quality-margin-first if extra quality is valuable.

6. Non-compliant products

For products below specification, define the quality shortfall:

$$s_i = q_{min} - q_i$$

Then rank non-compliant products by minimizing both shortfall and cost:

$$\text{minimize } s_i \text{ and minimize } c_i$$

This ensures the worst products are those with both large quality failure and high cost.

7. Recommended operational rule

Step	Rule
1	Separate compliant and non-compliant products using $q_i >= q_{min}$.
2	For compliant products, compute Pareto fronts using cost minimisation and quality maximisation.
3	Rank front 1 above front 2, front 2 above front 3, and so on.
4	Within each front, use a clear tie-breaker such as lower cost first.
5	Place all non-compliant products below compliant products.
6	For non-compliant products, rank by quality shortfall first, then cost.

Important note: This gives an ordinal decision ranking, not a smooth economic score. That is often an advantage: it avoids arbitrary weights and makes clear which products are objectively dominated.